41/100

CHE 313 Transport Phenomena III

Exam #3 (11/20/2025)

Name: Dr. ta Rodriguez

$x_F = 0.85$

1. (25pts) A simple steam distillation is being done in the apparatus. The organic feed is 85 mol% n-decane and 15% nonvolatile organics. The system is operated with liquid water in the still. A total condenser is used to produce pure n-decane from distillate (water+n-decane). Distillation is continued until the organic layer in the still is 5 mol% n-decane. F = 10 kmole. Pressure is 760 mm Hg.

$x_f = 0.05$ $F = 10$ kmol

1)(5pts) Draw a schematic of the steam distillation. Set up overall mass balance and component mass balance equations

2)(5pts) Derive the equation for W_f (amount liquid remaining in the pot)

3)(5pts) Calculate W_f and D_{total}

4)(10pts) Calculate W_f and D_{total} using the Rayleigh equation for simple batch distillation. Compare your answer with one from (3). (Hint: the produced n-decane is pure).

$x_{w,final}$

$$\ln\left(\frac{W_f}{F}\right) = -\int_{x_{w,final}}^{x_{w,initial}} \frac{dx_w}{x_D - x_w} = \ln\left(x_D - x_w\right)$$

2. (40 pts) 4.0 kmoles of a feed that is 60 mole % methanol and 40 mole % water is batch distilled in a system with a still pot (an equilibrium contact), a column that acts as one equilibrium stage (total two equilibrium contacts), a total condenser, and reflux to the column. p = 1.0 atm, the reflux ratio is L/D = 2.0, CMO is valid, and $x_{w,final}$= 0.08.

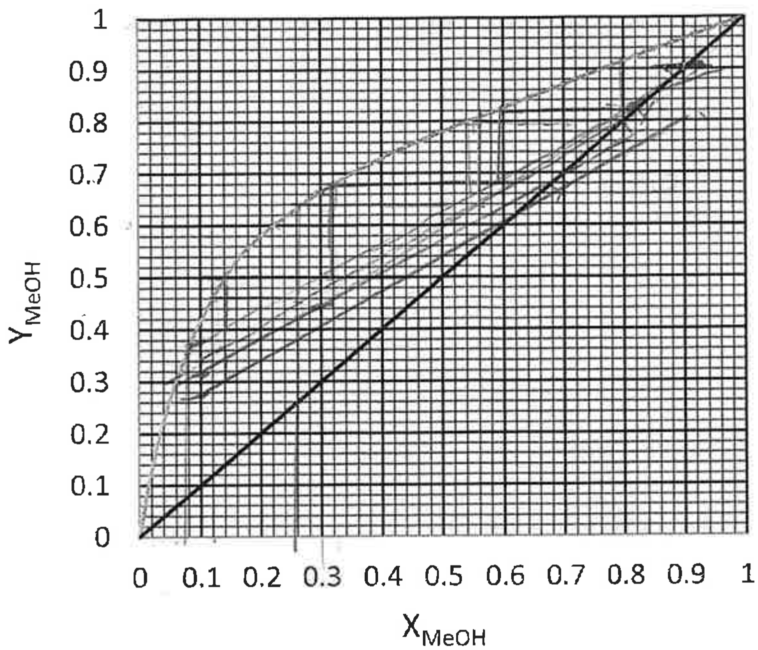
1) (5pts) Draw a schematic of the batch distillation. Define all parameters including F, W_f, V, L, and D_{Tot}. Set up overall mass balance and component mass balance.

2) (5pts) Set up equation for $x_{D,avg}$ from component mass balance equation.

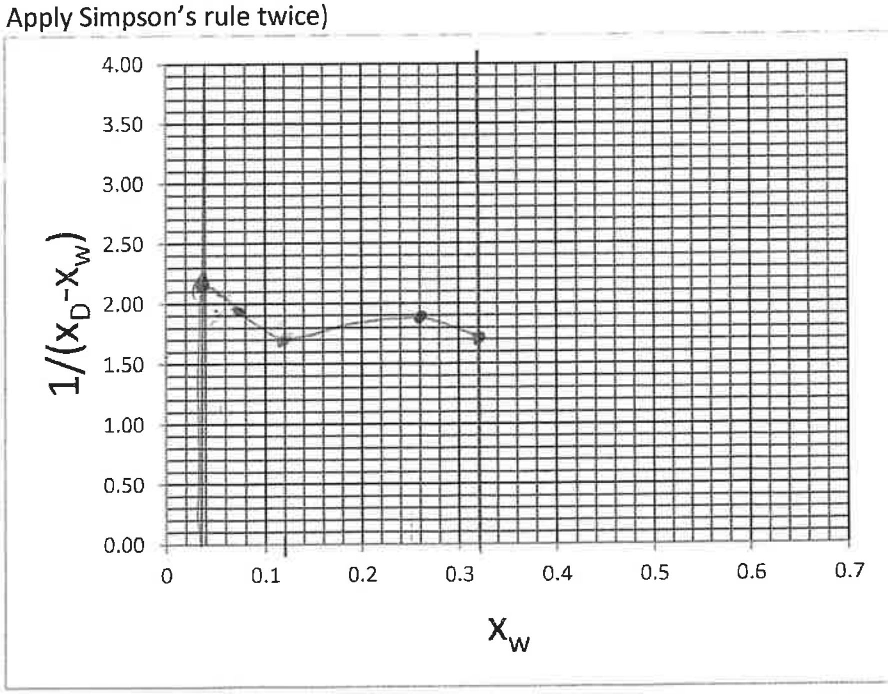
3) (5pts) Find the slope of operation equation, $y = (L/V)x + (1-L/V)x_D$

4) (10pts) Complete the table below using the VLE graph (draw OP lines and mark your x_w values on the graph).

	x_D	x_w	$x_D - x_w$	$1/(x_D - x_w)$	
*	0.88	~~0.32~~ 0.5	0.56	~~1.786~~	2.63
	0.8	0.258	0.542	1.84	
	0.70	0.12	0.58	1.72	
*	0.60	0.08 ~~0.065~~	0.535	1.869	
	0.50	0.048	0.450	2.375	



5) (5pts) Build the integration curve for x_w vs. $1/(x_D - x_w)$ and calculate the area under the curve (Hint: Apply Simpson's rule twice)



6) 5) (10pts) Find W_{final}, D_{total}, and $X_{D,avg}$.

Simpson's rule:

$$Area = \frac{x_{Feed} - x_{final}}{6}\left[\frac{1}{x_D - x_w}\bigg|_{x_{final}} + \frac{4}{x_D - x_w}\bigg|_{x_w,middle} + \frac{1}{x_D - x_w}\bigg|_{x_{Feed}}\right]$$

Rayleigh equation: $W_{final} = F\ exp\left(-\int_{x_{final}}^{x_{Feed}} \frac{dx_w}{(x_D - x_w)}\right)$

L =

3. **(35 pts) We are absorbing acetone at 25C into water. 1000.0 kmol/hr of air containing 100 ppm of acetone and 2.0 atm is to be processed. Entering water from stripper already contains 0.1 ppm of acetone. Recovery of acetone in the water is 80%. At 25 °C, equilibrium relation is $y = 105.6x$.**

   a) (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Set up overall mass balance and component mass balance around stage j.

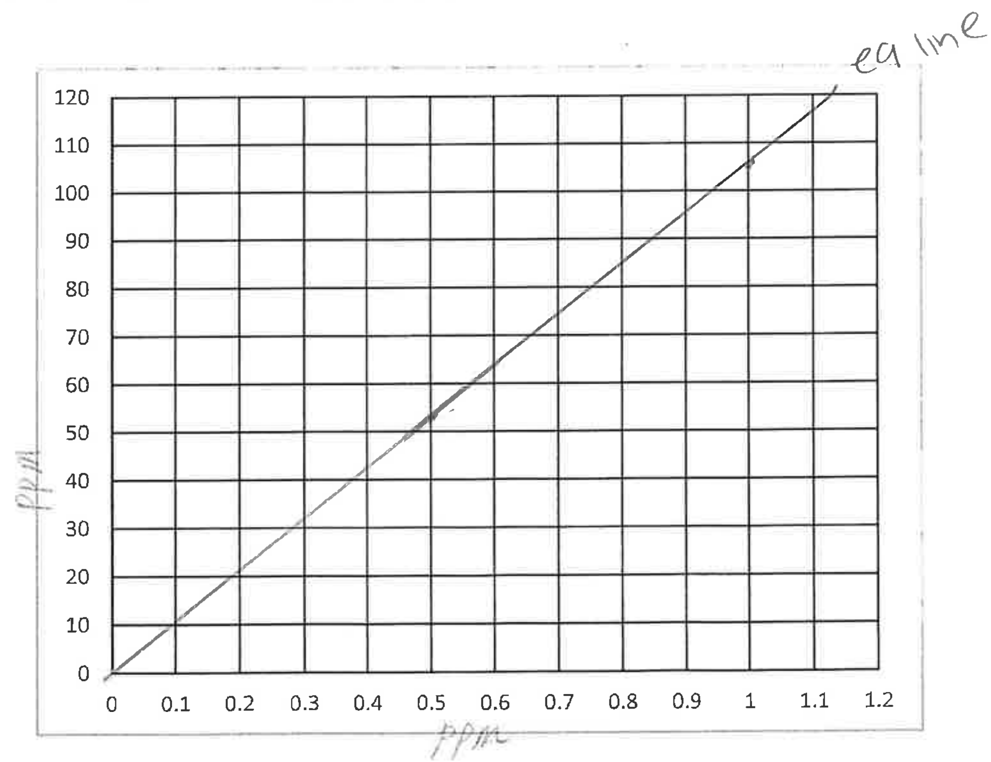
   b) (5 pts) Derive operating equation using component mass balance. 0-20

   c) (5 pts) Find (x_0, y_1) and (x_N^*, y_{N+1}). x_N^* is x in equilibrium with y_{N+1}.

   d) (5 pts) Plot equilibrium and minimum operating lines.

   e) (5 pts) When $L/V = 1.59\ (L/V)_{min}$, find L/V, x_N, and y-intercept of operating equation

   f) (10 pts) Plot operating line. Find the number of equilibrium stages.



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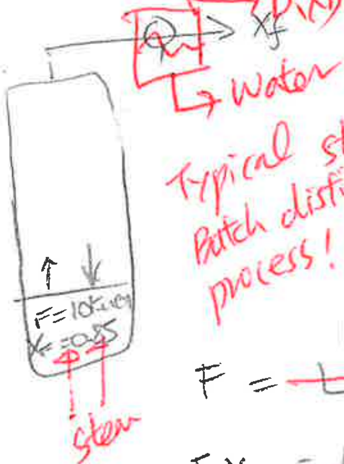
1)



Typical steam Batch distillation process!

$x_F = 0.85$

$x_f = 0.05$

$F = 10$ kmol

find W_f & D_{total}

1) 2/5

$F = \cancel{L + V}$ $W_f + D_{Total}$

$F x_F = \cancel{L x_f + V x_D}$ $W_f \cdot x_f + D \cdot x_{D,avg}$

x	x_w	$\frac{1}{x_w}$
0.85	0.15	6.66
0.45	0.55	1.81
0.05	0.95	1.05

2)

$W_f = F e^{-\text{Area}}$

2) 1/5

Simpsons Rule

3) 2/5

$W_f = F\left(\frac{x_D - x_F}{x_D - x_f}\right)$

$W_f = \cancel{1.413}$ kmol

1.57 kcal

$D_{total} = 8.43$ "

$\frac{W x_F - W_f}{6}\left[W_F + 4(x_{mid}) + W_f \right]$

3

$F = W_f + D_{total}$

$10\text{kmol} = 1.413 + D_{total}$

$D_{total} = 8.57$ kmol

$\frac{0.80}{6}\left[6.66 + 4(1.81) + (1.05) \right]$

$0.13\left[14.95 \right]$

$A = 1.94$

4) $\ln\left(\frac{W_f}{F}\right) = -\int$

$\frac{W_f}{F} = e^{-\int_{x_f}^{x_w} \frac{dx_w}{x_D - x_w}}$

$-\ln \frac{dx_w}{x_D - x_w} \Big|_{x_f}^{x_w}$

$W_f = \frac{\cancel{e}}{F}$

4) 2/10

$W_f \quad \frac{Dx_w}{}$

$-\int_{0.05}^{0.85} \frac{dx_w}{1 - x_w} = \ln(1 - x_w)\Big|_{0.05}^{0.85}$

$\Downarrow$

$\frac{W_f}{F} = \exp[\ln 0.15 - \ln 0.95]$

$= 0.158$

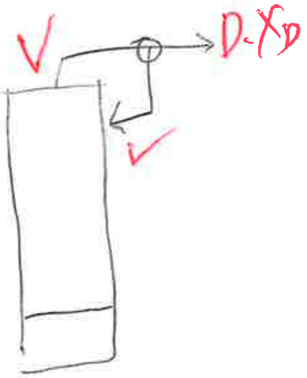
2) $F = 4.0$ kmol

$x_F = 0.60$

$F = V + L$

$F x_F =$

1)



$\frac{L}{D} = 2.0$

1) 2/5

$x_{w,final} = 0.08$

$F = W_f + D_{tot}$ ← overall M.B

$F x_F = W_f x_f + D \cdot x_D$ ← comp "

2) $x_{D\,avg} = \frac{F x_F - W_f x_f}{D_{tot}}$

2) 5/5

$\frac{L}{V} = \frac{L/D}{1 + \frac{L}{D}} = 0.66$

3) $y = \left(\frac{L}{V}\right) x + \left(1 - \frac{L}{V}\right) x_D$

3) 5/5

$y = 0.66x + (1 - 0.66) x_D$

$y = 0.66x + 0.34 x_D$

4) 8/10

5) 2/5

4)

x_D
0.88

$y = 0.66x + 0.34(x_D)$

$y = 0.66x + 0.2992$

0.80 $\quad y = 0.66x + 0.272$

0.338

Area ≅ 0.86

0.70 $\quad y = 0.66x + 0.34(x_D)$

$y = 0.66x + 0.238$

0.6 $\quad y = 0.66x + 0.34(0.6)$

x_D
0.5 $y = 0.66x + 0.17$

$\frac{x_D}{0.6}$ $y = 0.66x + 0.204$

5) $A = \frac{\cancel{0.048} - 0.12}{\cancel{0.5} \quad 6}\left[\cancel{2.375 + 4(1.869) + 1.72}\right]$

$= 0.1388$

$A = \frac{\cancel{0.12 - 0.32}}{6}\left[\cancel{1.72 + 4(1.84) + 1.786}\right]$

0.125 − 0.08

6) 6/10

$A = 0.3622$

$A_{total} = \cancel{0.501}$ 0.86

-0.501

$W_{final} = 4.0e$

D total ?

$F = W_f + D_{tot}$

$W_f = \cancel{2.412}$ 1.70 kmol

$D_{tot} = F - W_f$

$x_{D,Avg} = \frac{4.0(0.6) - \cancel{2.412(\quad)} \ 1.1 \times 0.08}{\cancel{1.58} \ 2.77}$

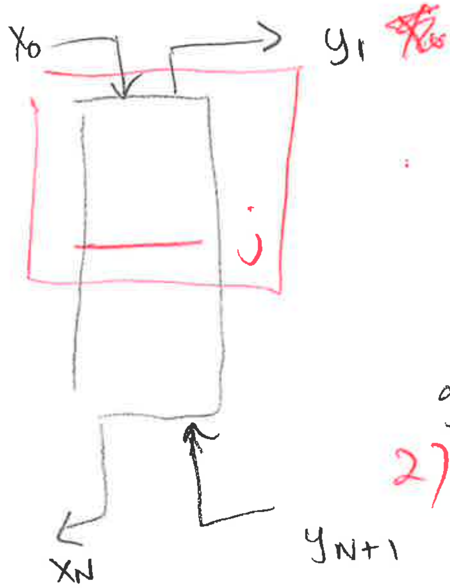
$D_{tot} = \cancel{1.58}$ kmol
2.30

≈ 0.83

3) $F = 1000 \frac{Kmol}{hr}$ $P = 2.00\ atm$

$y_{in} = V = 100\ ppm$

$y_{out} = 0.30\ x$ $x_{in} = L_{in} = 0.1\ ppm$ overall M.B = ?



b) $Lx_0 + \cancel{G}y_{N+1} = \cancel{G}y_1 + Lx_N$

$y_{out} = 0.3(0.1\ ppm)$

y_1 $y_1 = 0.03$

given = $y = \frac{H}{P_{tot}} x = 105.6x$

2) op: $y = (\frac{L}{V})x + [y_1 - (\frac{L}{V})x_0]$

1) 3/5

2) 0/5

3) 0/5

4) 2/5

$y_{in} V = y_{out} V + x_{out} L$

$\frac{L}{G_{min}} \left(\frac{y_{in} - y_{out}}{x^* - 0} \right) \Rightarrow \frac{L}{G} = (105.6) \frac{L}{G_{min}}$

$y_{in} V = y_{out} V + x_{out} L$

$x_{out} = (y_{in} - y_{out}) \frac{V}{L}$

$y = \frac{L}{V} x_{in} = \frac{L}{V}(x_{in} - x_{out}$